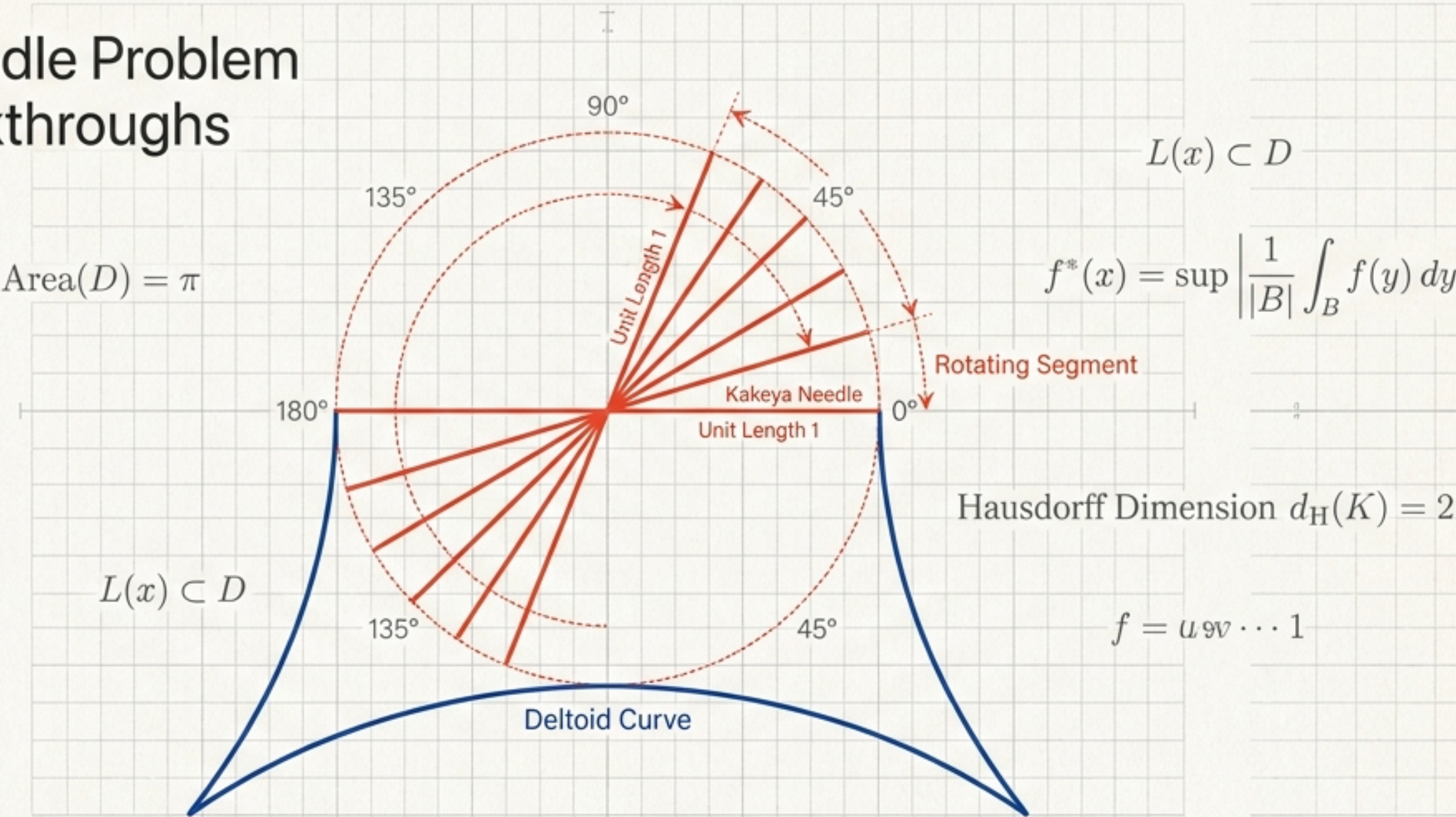


The Kakeya Conjectures

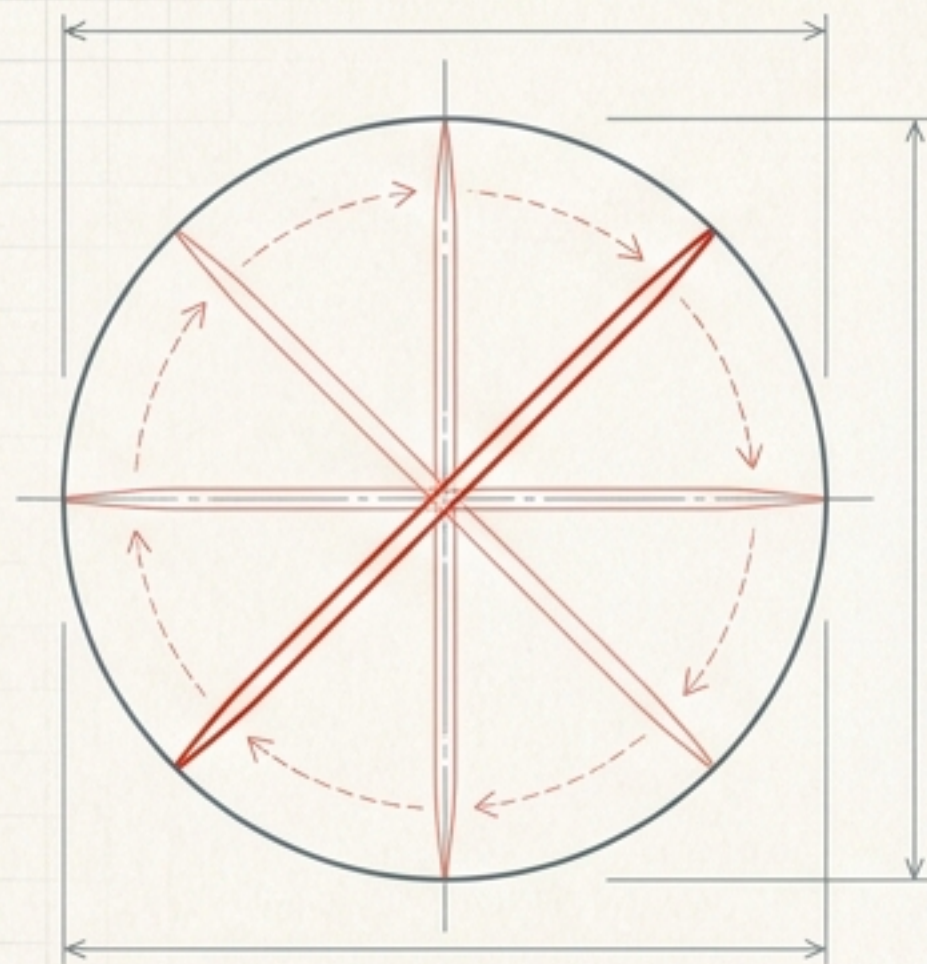
From a 1917 Needle Problem to 2026 AI Breakthroughs



An analysis of geometric measure, maximal functions, and the automated frontier.

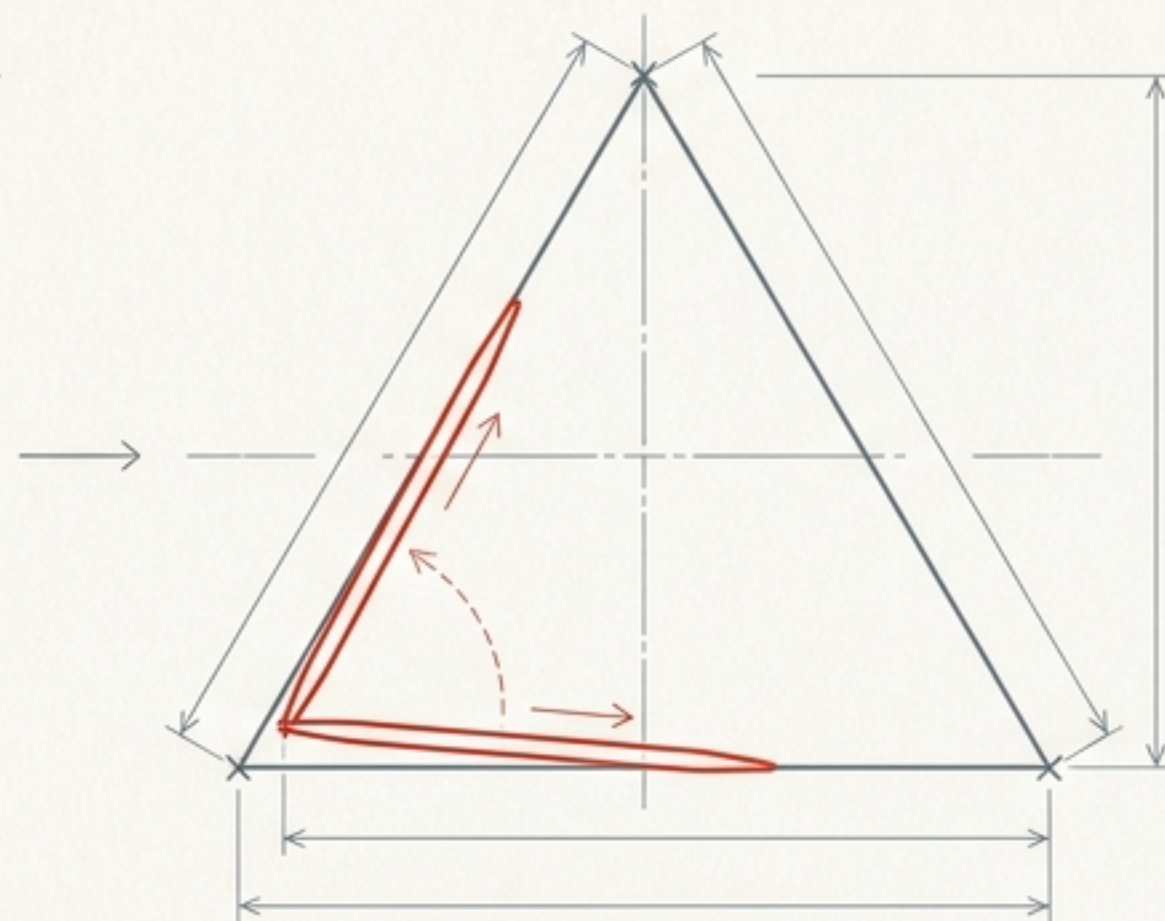
The 1917 Physical Puzzle

Sōichi Takeya originally asked: What is the smallest possible area in which a line segment of length 1 can be continuously rotated a full 360 degrees?



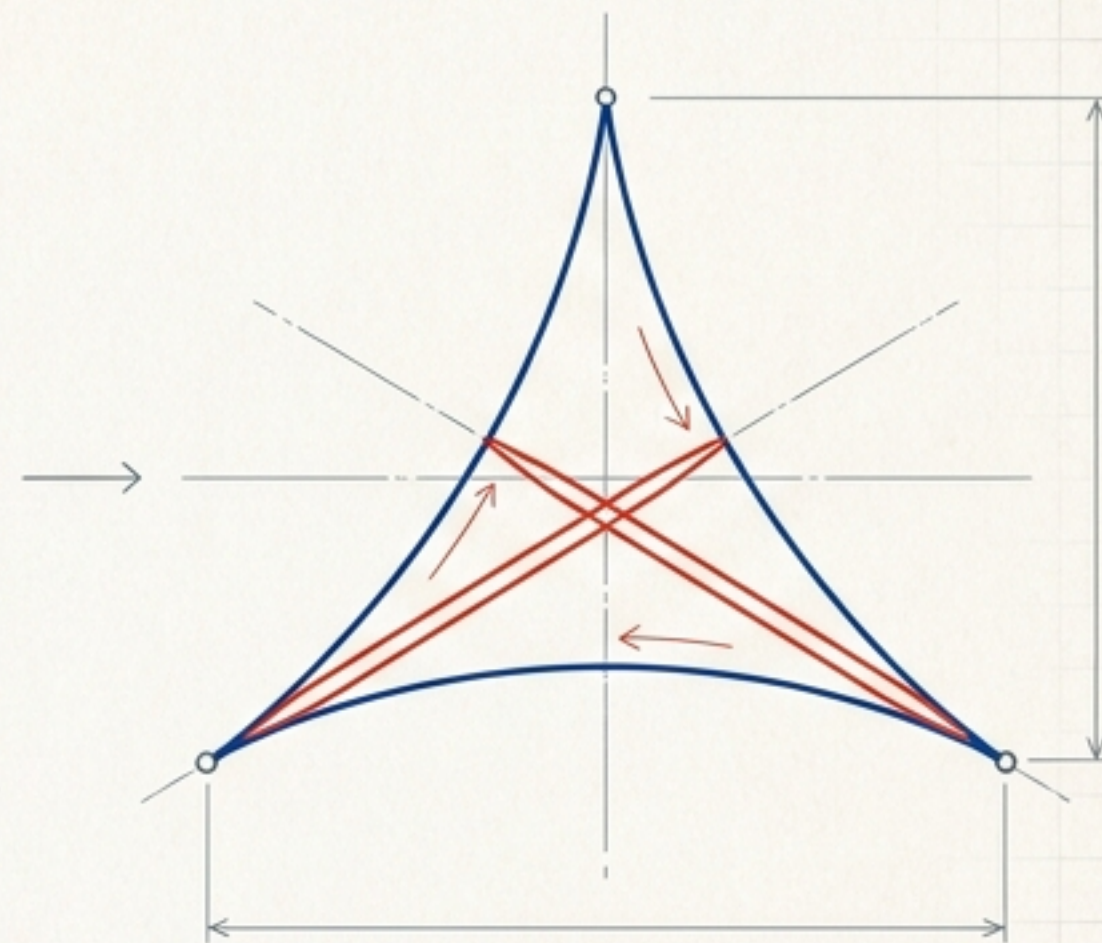
Area: $\frac{\pi}{4} \approx 0.785$

A unit needle rotates easily.



Area: $1/\sqrt{3} \approx 0.577$

Proved by Gyula Pál, 1920.

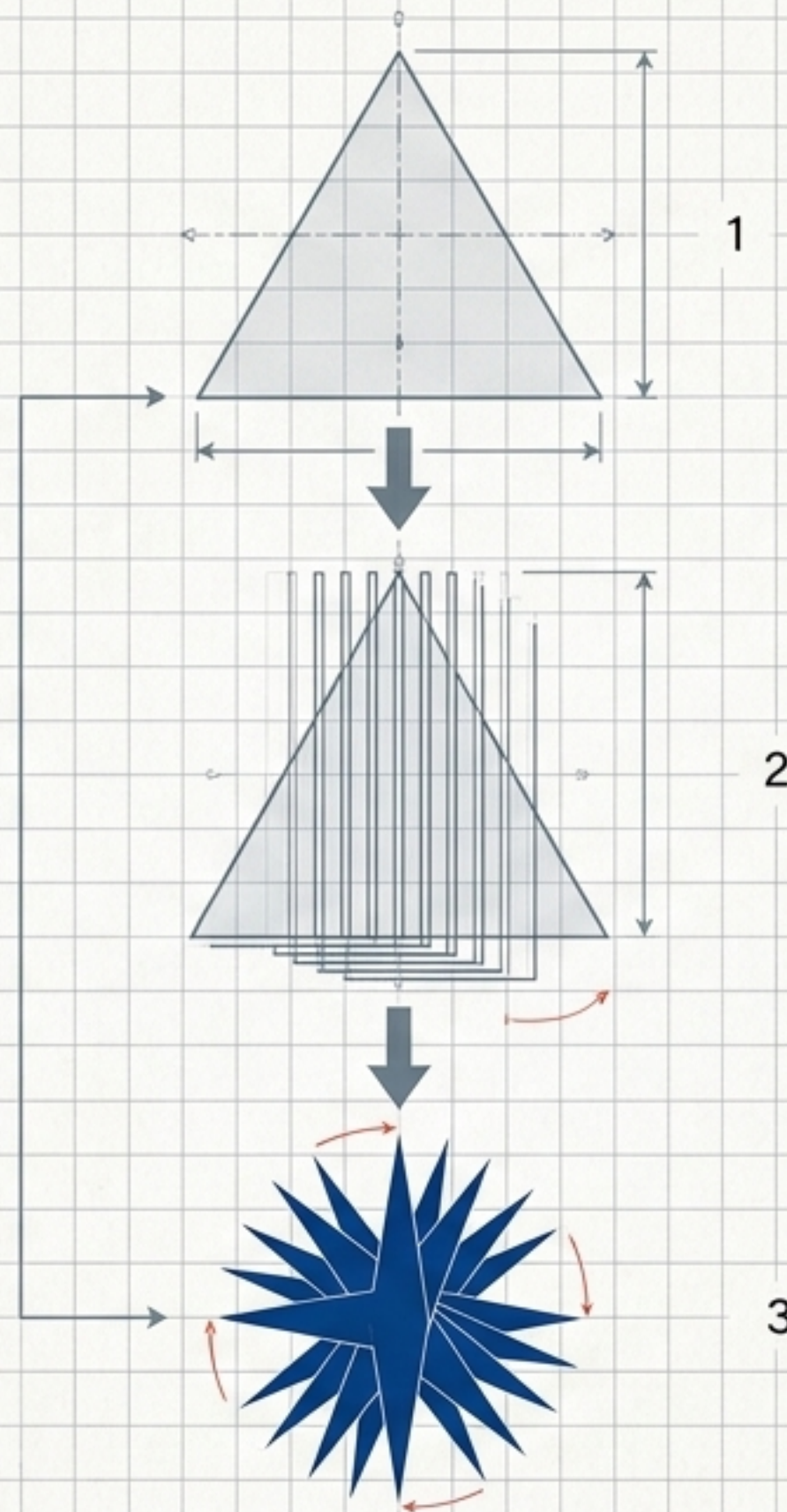


Area: $\frac{\pi}{8} \approx 0.392$

The initial assumption was the deltoid. It was incorrect.

The Paradox of Area Zero

In 1920, Abram Besicovitch proved there is no minimum area. A needle can be fully rotated in a space with an area arbitrarily close to zero.



By dividing a triangle and overlapping the pieces, the area shrinks to zero, yet every rotational direction is preserved.

The Kakeya Set Conjecture

- A Kakeya Set in $\mathbb{R}^n$ is any set containing a unit line segment in every possible direction.
- The Conjecture: Every such set must have a Hausdorff dimension exactly equal to the ambient space n .

1971



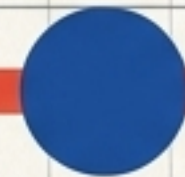
Davies proves the
2D case.

1995



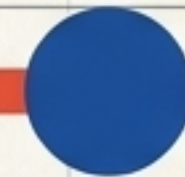
Wolff establishes
 $(n+2)/2$ dimensional
bounds.

2000s



Bourgain, Katz, and
Tao apply arithmetic
combinatorics.

2025

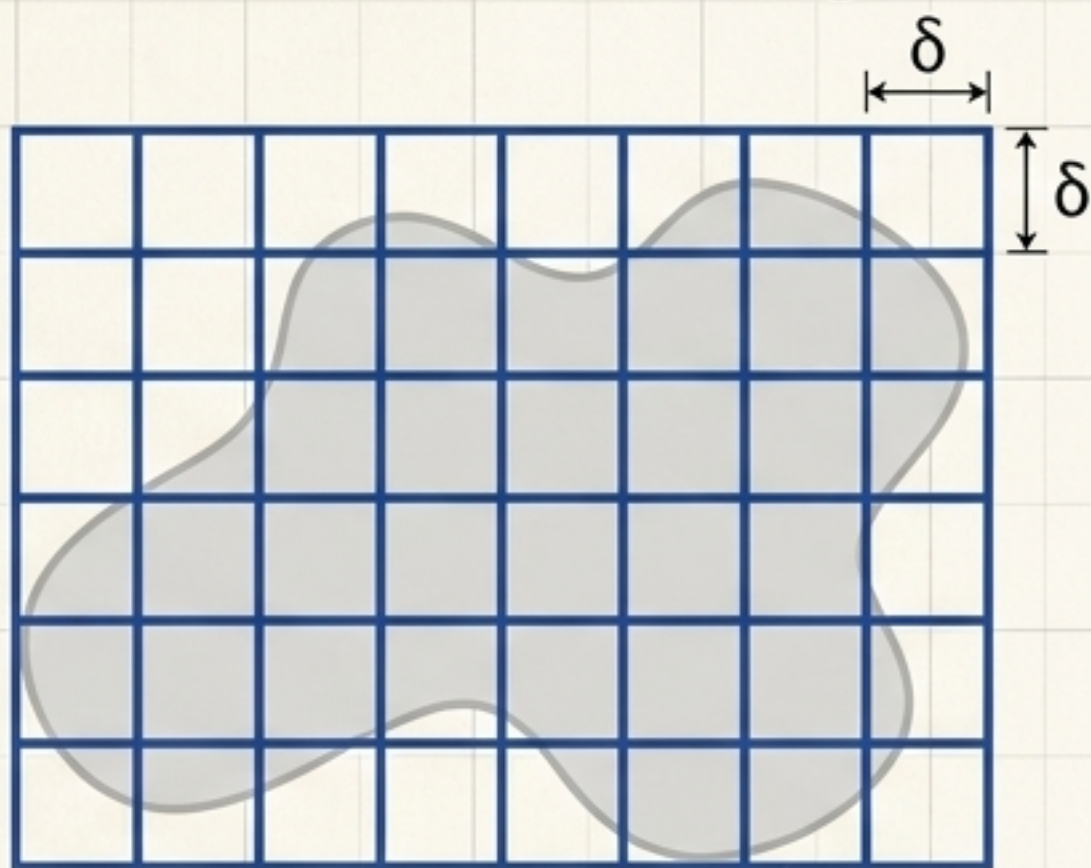


Wang & Zahl prove
the 3-dimensional
case.

Building the Finer Ruler

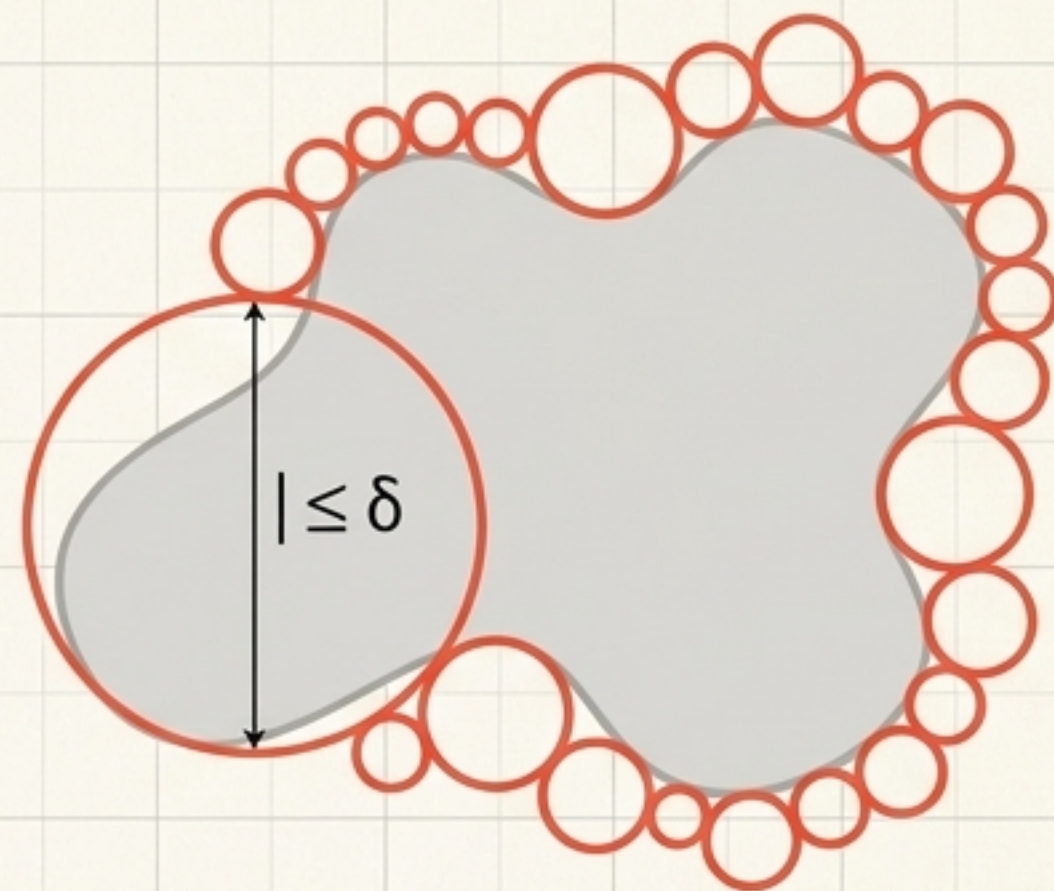
How do we measure size when area is zero? Proving **full Hausdorff dimension** is the ultimate test.

Minkowski Dimension: The Rigid Ruler



Counts the number of fixed grid cubes of side δ needed to cover the set. Easier to calculate, but assumes uniformity.

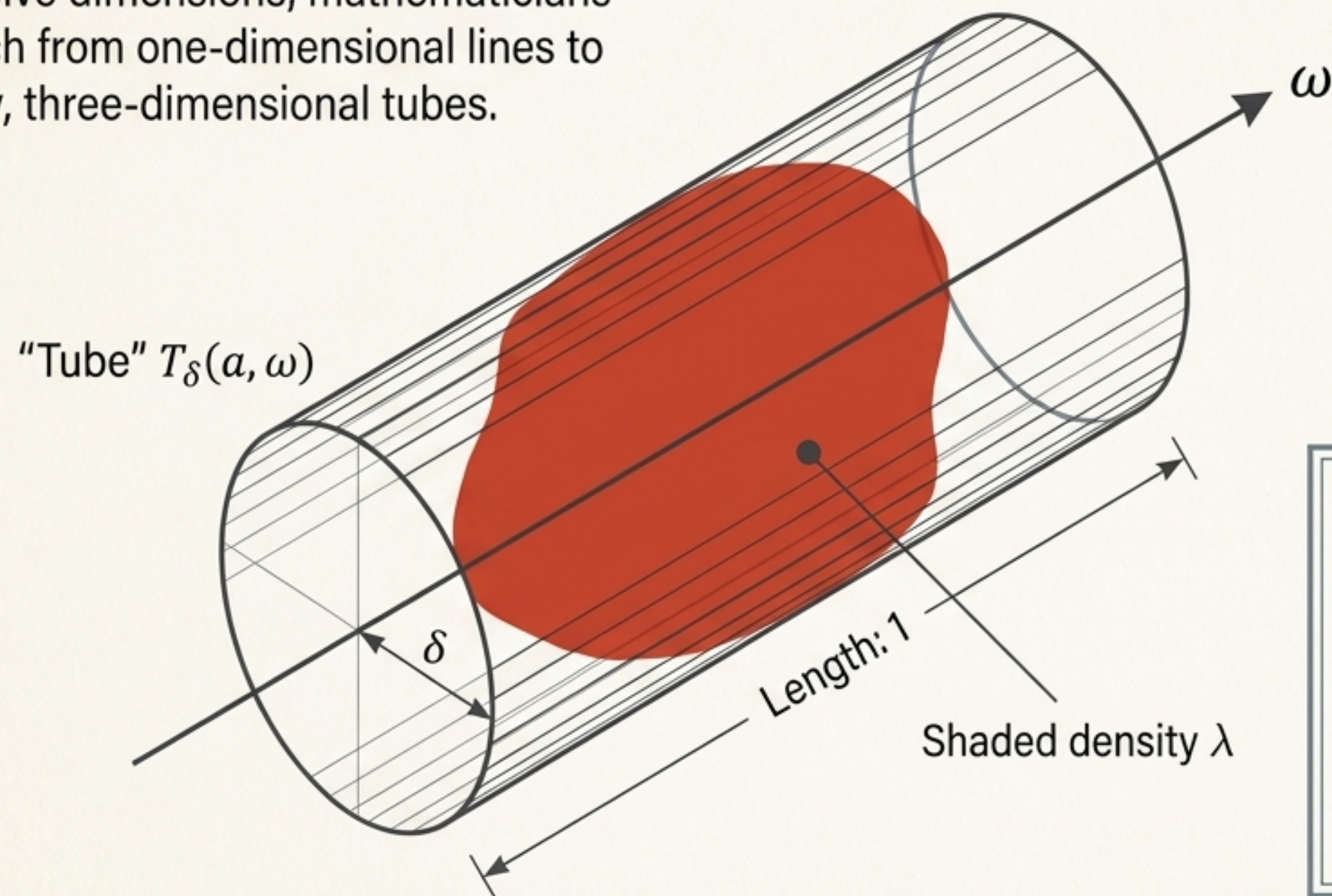
Hausdorff Dimension: The Flexible Ruler



Allows covers of varying sizes up to δ . It is the strictly stronger, finer measure of a set's true size.

The Shift to Analysis: Tubes and Shadings

To solve dimensions, mathematicians switch from one-dimensional lines to fuzzy, three-dimensional tubes.

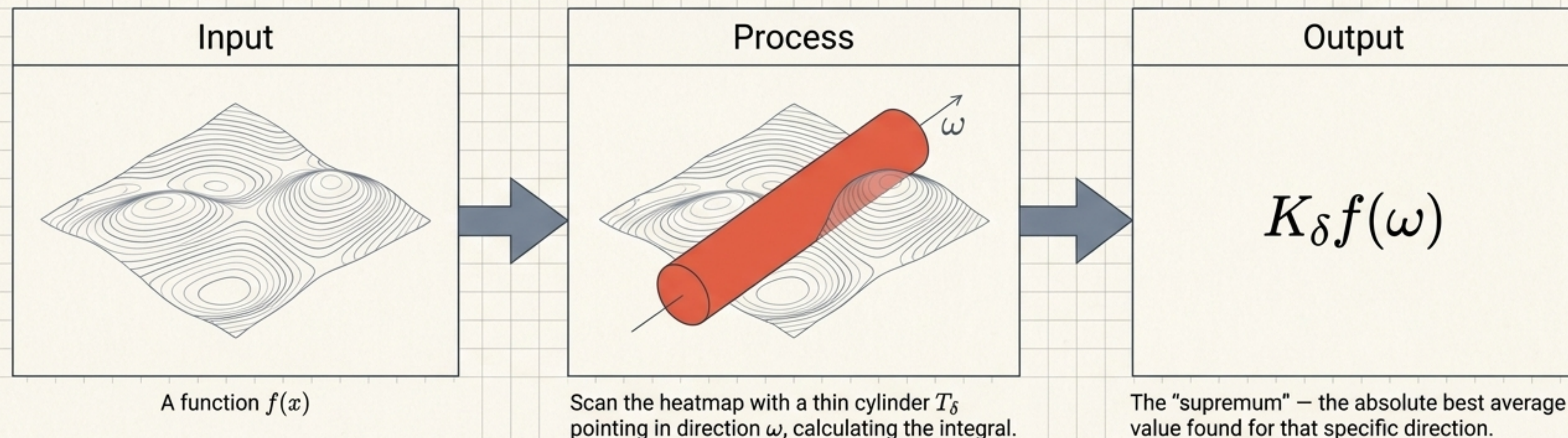


If tubes of radius δ are shaded with a density λ , the volume of their overlapping union must be proportionally massive.

$$\text{Union Volume} \geq \lambda^n \times \text{Total Volume}$$

The Kakeya Maximal Function

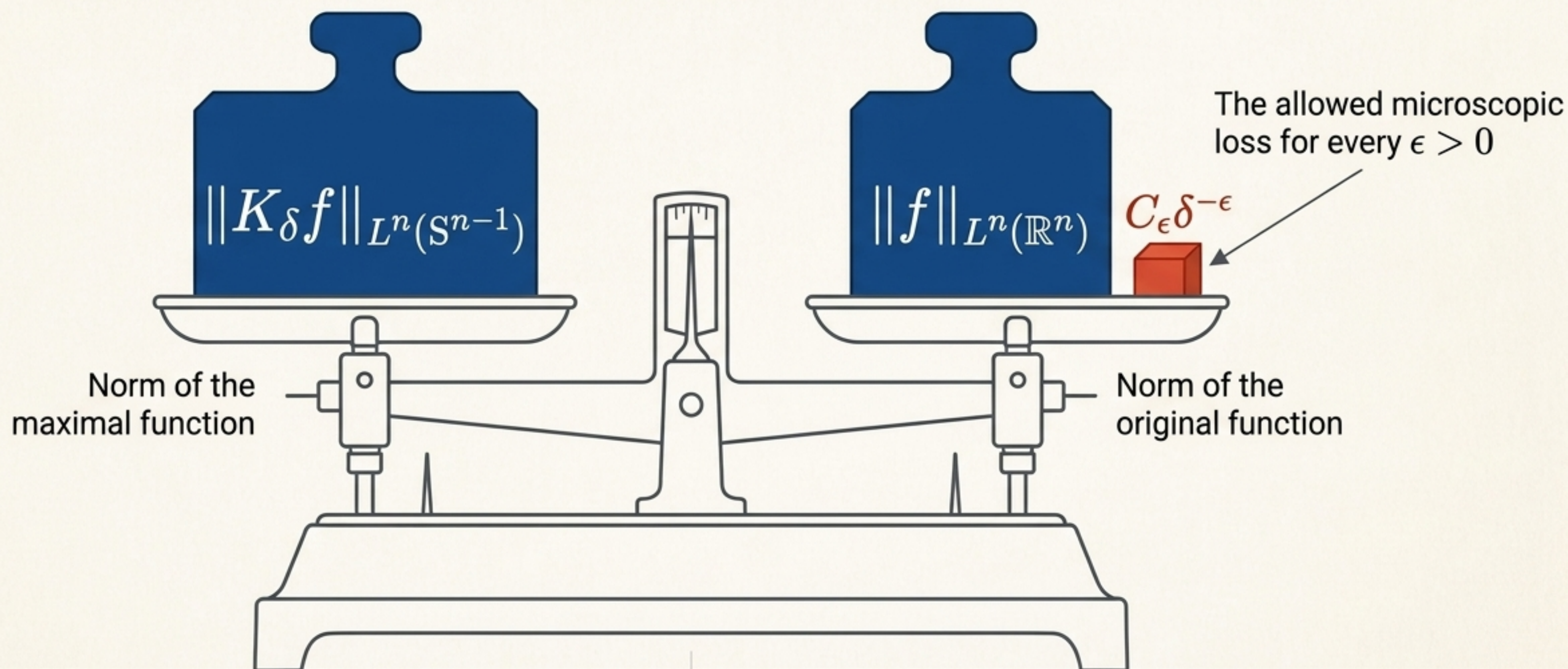
Instead of looking at static sets, we measure how a function $|f|$ averages out over thin tubes.



$$K_\delta f(\omega) = \underbrace{\sup_a}_{\text{Supremum over all positions}} \underbrace{\frac{1}{|T_\delta|}}_{\text{Normalization by tube volume}} \underbrace{\int_{T_\delta(a, \omega)} |f(x)| dx}_{\text{Integral over tube}}$$

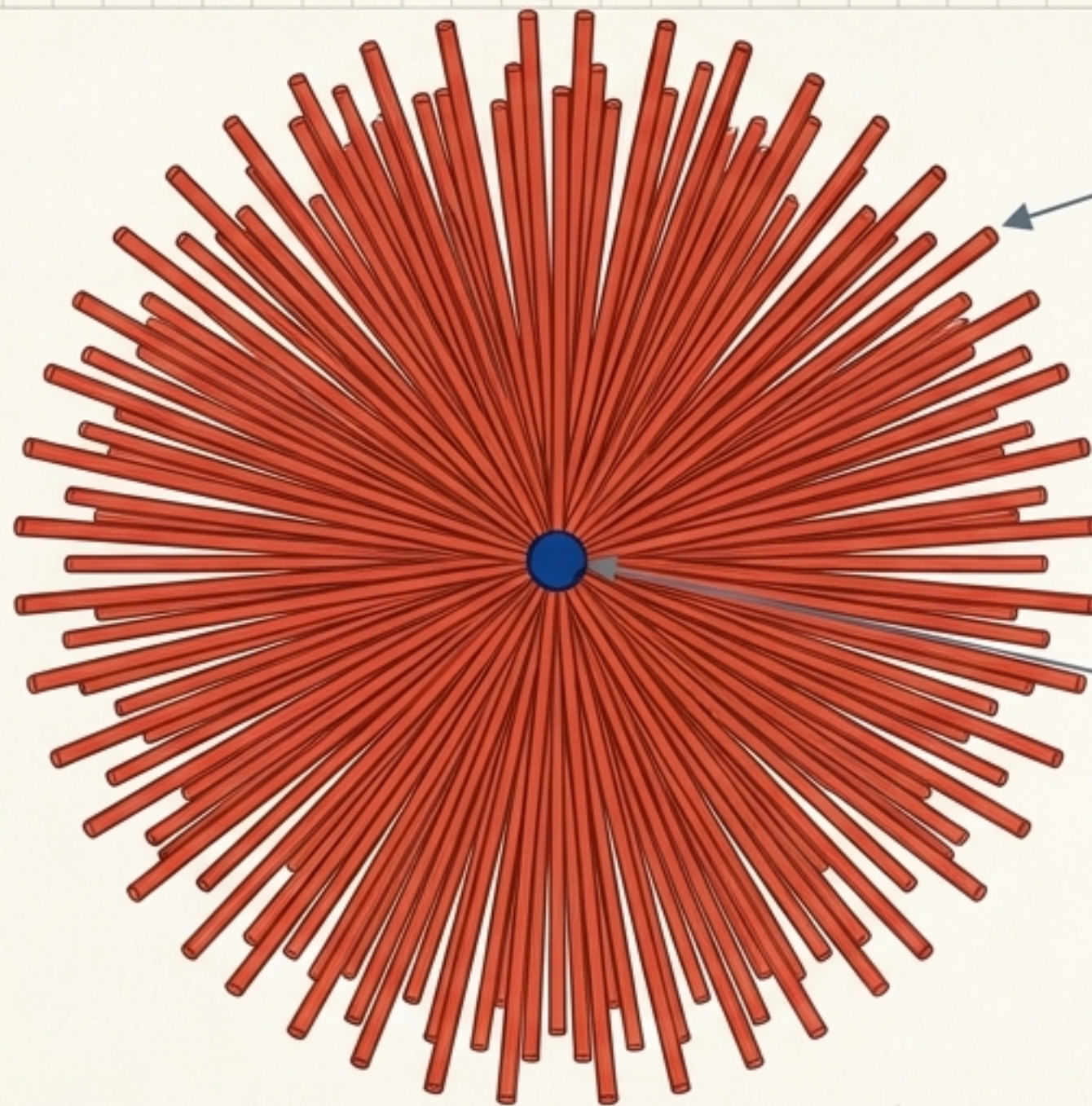
The Maximal Conjecture

It demands that tube overlaps cannot conspire to hide the function's mass.
We want an L^n bound with only a microscopic loss factor.



The Bush Example: A Forced Geometric Traffic Jam

Why must the bound specifically be L^3 in $\mathbb{R}^3$? A physical bottleneck forces the math.



1 Dispersed Volume

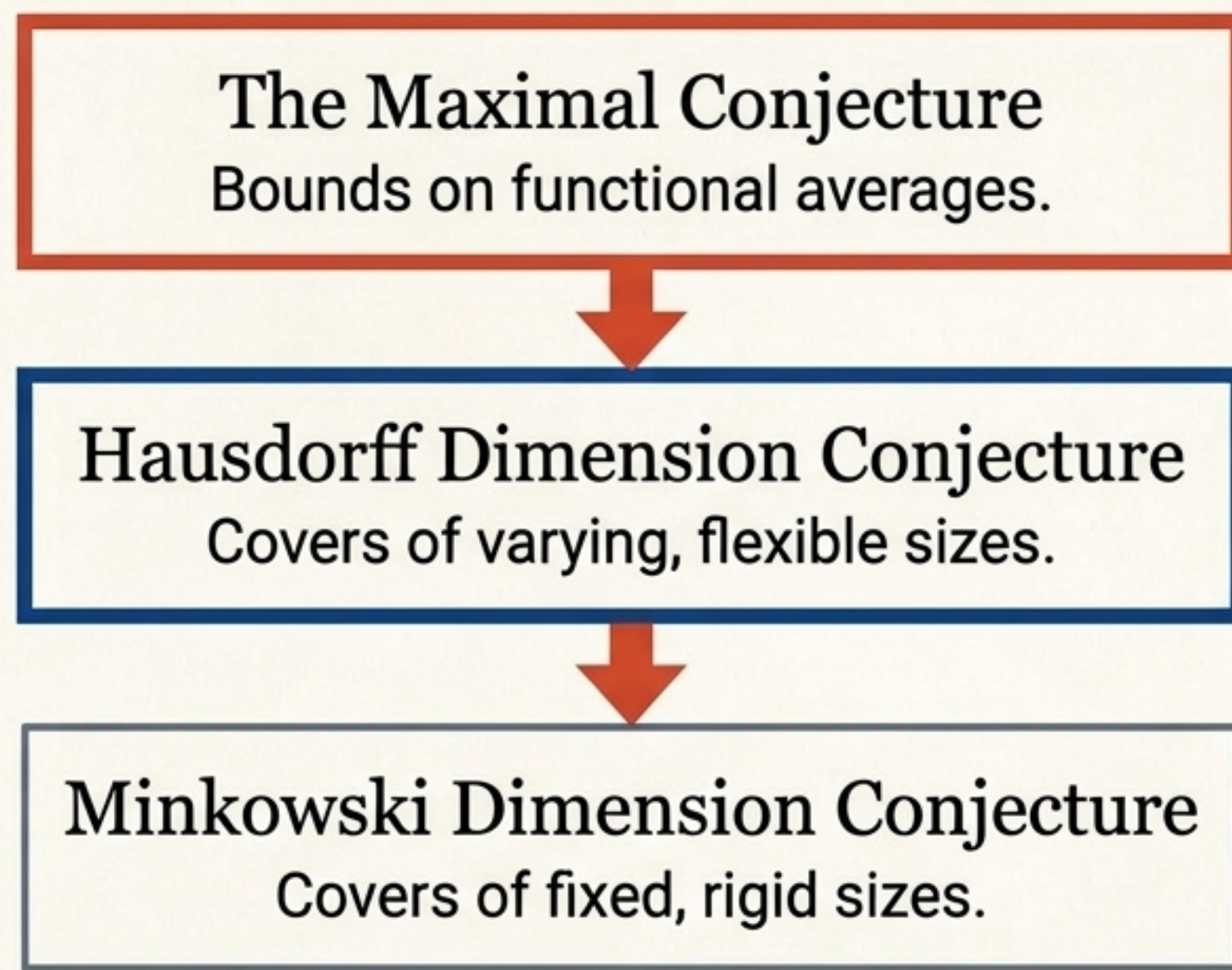
At the edges, tubes are separated and volume is maximized.

2 Central Bottleneck

If you force all tubes to pass through one tiny central ball, the math hits a physical limit. This "bush" proves that L^3 bounds (and λ^3 volume bounds) are absolutely forced in 3D space.

The Hierarchy of Truth

Proving the Maximal Conjecture is the ultimate prize because it triggers a cascade of geometric truths.



Maximal bounds inherently guarantee full Hausdorff dimension, which in turn automatically guarantees full Minkowski dimension.

Enter the Machines: September 2026

Over two days, twin AI-written preprints claimed massive leaps on the Kakeya problem, resolving decades-old bottlenecks using complex structural geometries.

OpenAI Preprint

The Kakeya maximal conjecture
in three dimensions

Sep 23, 2026

OpenAI Preprint

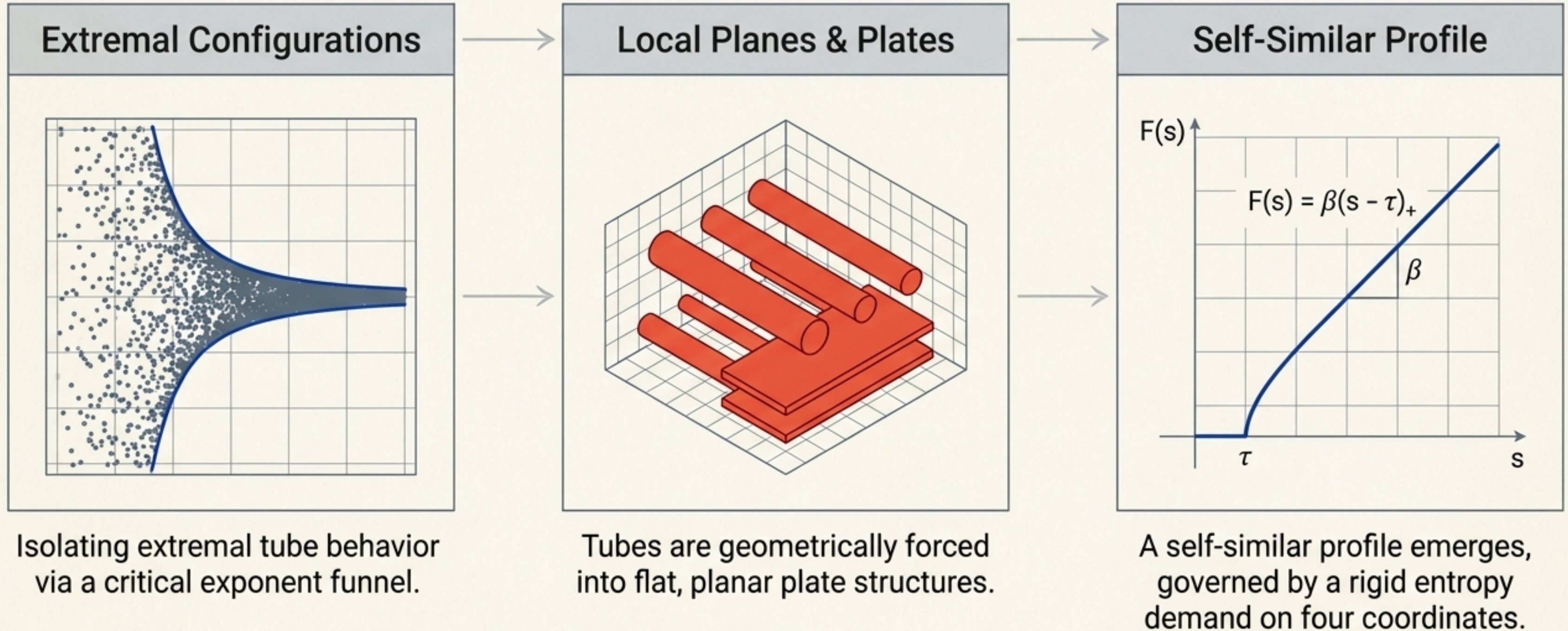
Every Four-Dimensional
Kakeya Set Has Full Hausdorff
Hausdorff Dimension

Sep 24, 2026

WARNING: Unreviewed preprints.
Awaiting human peer-review and
Lean formalization.

Paper 1: The 3D Maximal Conjecture Solved

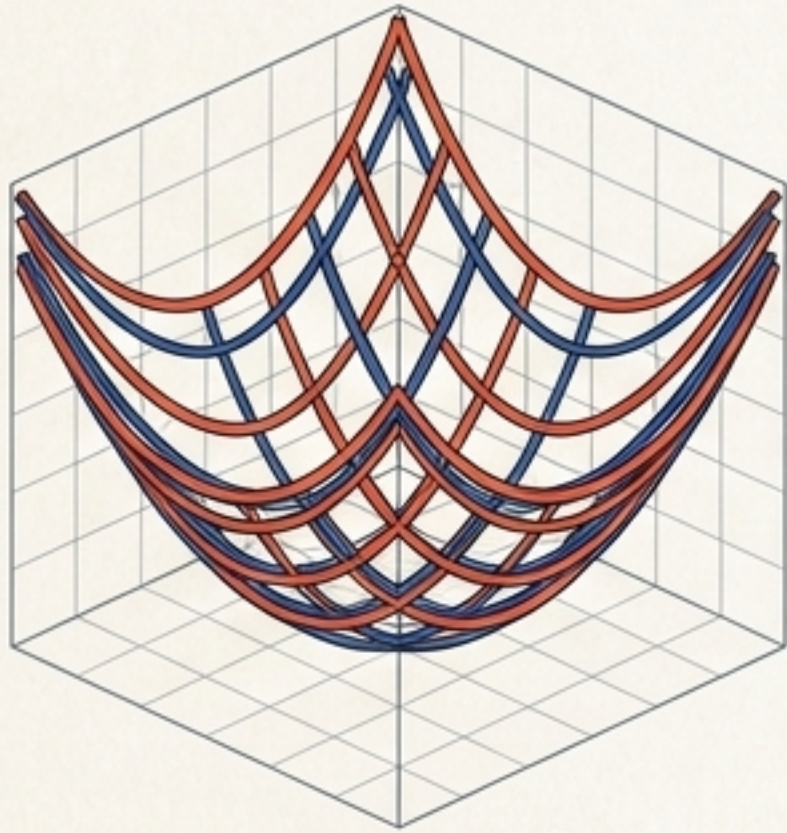
The AI proves the 3D maximal conjecture by studying extremal configurations, identifying a profile whose entropy demand contradicts projection estimates.



Paper 2: 4D Hausdorff Dimension Solved

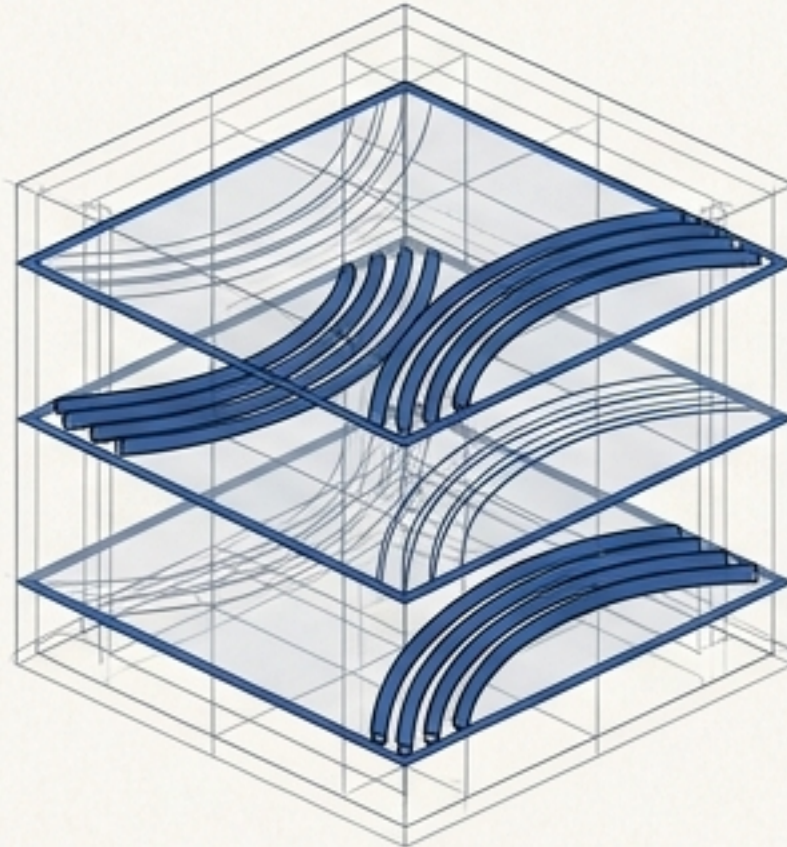
Every subset of $\mathbb{R}^4$ containing a unit segment in every direction has dimension 4.
The proof maps tube concentrations across three distinct narrowness regimes.

$$\ell = 0$$



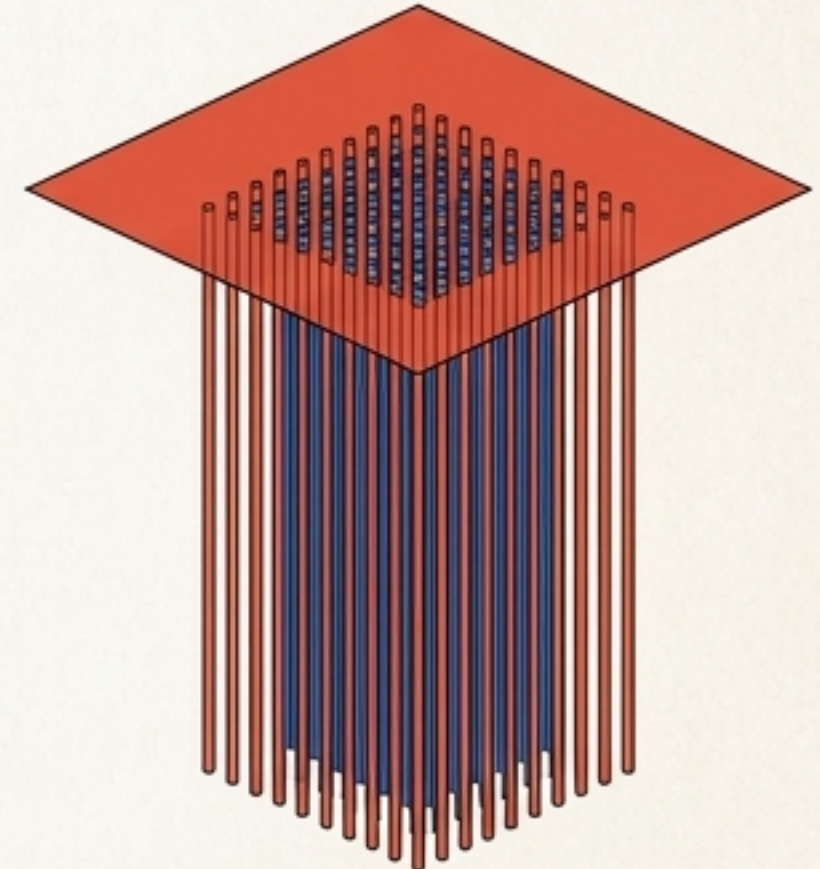
Regime 1: Tubes form a polynomial potential. The field is switched to become approximately conservative.

$$0 < \ell < \infty$$



Regime 2: Tubes are trapped in a horizontal model, tracked via nested quadratic charts.

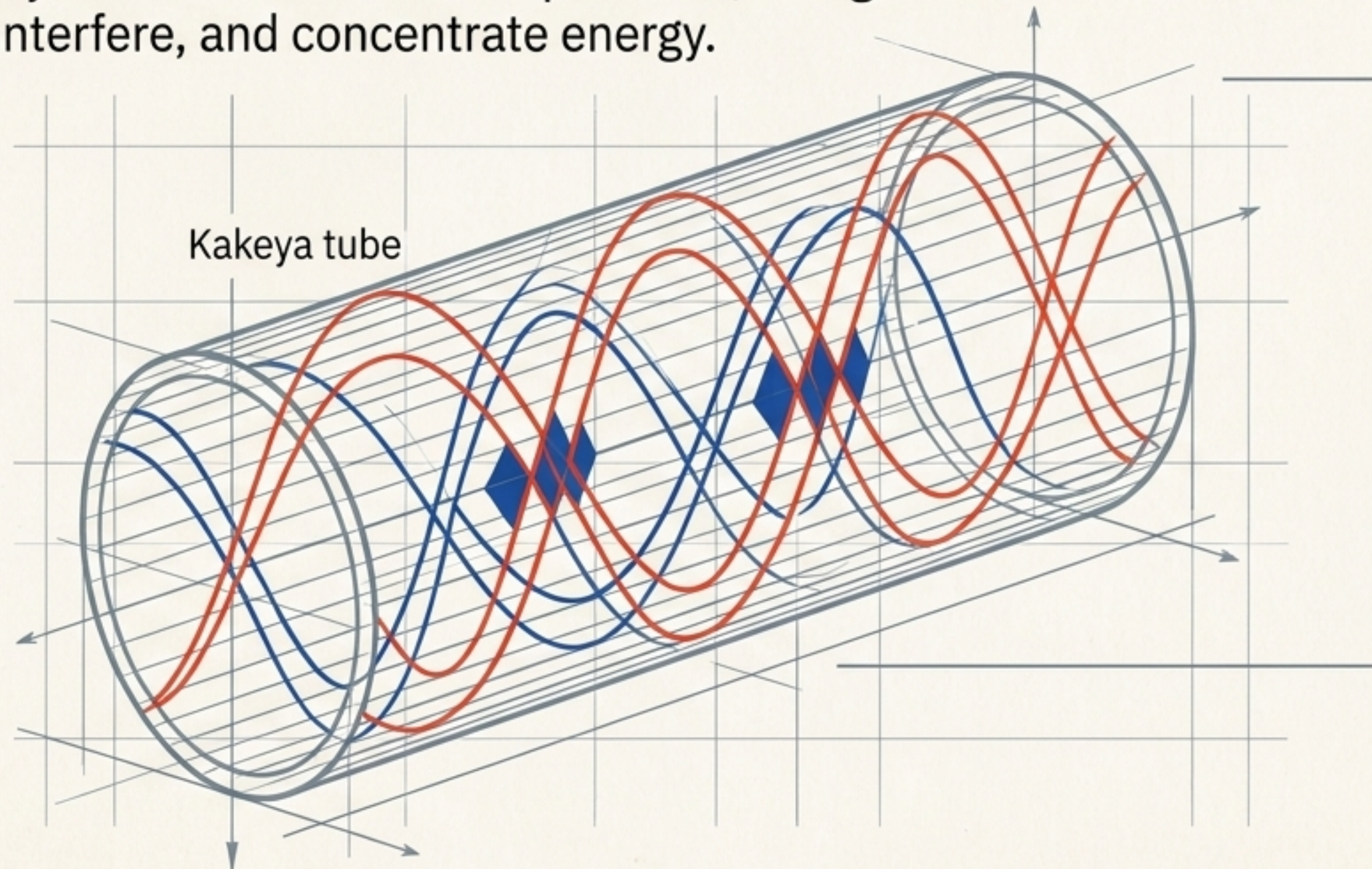
$$\ell = \infty$$



Regime 3: Full-time localization and scalar projections mathematically crush the tubes.

Why Does This Matter? The Deep Links

The Kakeya problem governs Fourier restriction. Geometric "traffic jams" dictate how wave packets (like light or sound) interact, interfere, and concentrate energy.



From Pure Geometry to Physics: By understanding how geometric tubes overlap in multi-dimensional space, mathematicians can predict wave interference patterns.

The New Frontier

A 100-year-old needle problem has been pushed into a new era by automated synthesis. What remains for human and machine verification?

3D Maximal Conjecture	✓	4D Hausdorff Dimension	✓
SOLVED (AI, 2026)		SOLVED (AI, 2026)	
4D Maximal Conjecture	✗	5D+ Conjectures	✗
OPEN		OPEN	

The 4D Maximal problem and dimensions five and higher remain entirely open.